

KHIEM TON

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EDUCATION

New Jersey Institute of Technology

September 2024 - Present

Ph.D. in Data Science

Focus: LLMs, VLMs, Post-Training (SFT & RL), Agentic Systems, AI for Software Engineering, Security, Privacy

TECHNICAL SKILLS

- **Machine Learning:** LLMs, VLMs, Diffusion Models, NLP, RL, Distributed Training, GNN, Federated Learning, Distillation.
- **Frameworks:** Pytorch, Transformers, verl, vllm, Ray, Triton, DeepSpeed, Accelerate, Pytorch Geometric.
- **Infrastructures & Tools:** CUDA, Linux, Git, AWS, Docker, SQL, Snowflake
- **Programming Languages:** Python, Java, JavaScript, C/C++

EXPERIENCE

Qualcomm | *Research Collaborator*

January 2026 – Present

- **Mobile Agentic Systems:** Co-designed an agentic system that leverages VLM to predict and perform user intentions on mobile devices. Developed an end-to-end mobile application to collect real-world usage data.

AT&T | *Research Collaborator*

November 2025 – Present

- **Cellular Network Maintenance Scheduling:** Designed a graph foundation model to predict cellular sites maintenance, optimizing for both operational efficiency and user experience. Processed and analyzed large-scale historical LTE and 5G network data to train and evaluate scheduling systems.

New Jersey Institute of Technology | *Research Assistant*

August 2024 – Present

- **RL for Multimodal Reasoning:** Designed RL experiments to improve VLM math problem solving, yielding gains across math, science, and general reasoning tasks.
- **RL for LLM vulnerability repair:** Designed VRPO, a RL framework leveraging formal verification to improve LLM capabilities on generating functionality correct and secure patches by 2×. Engineered multi-GPUs and multi-node distributed GRPO training pipeline using verl, DeepSpeed on an HPC cluster, achieving 4× speedup. Curated a verifiable code dataset of security and functionality tests for training and evaluating using a Code Agent System.
- **Privacy-Respecting LLM System:** Co-developed NOIR, a distributed LLM system using differential privacy to prevent 100% prompt and response reconstruction attacks. Reduced NOIR's inference latency by 3× by creating a distributed inference system using Ray, modified vllm with CUDA and high-throughput gRPC packet transfer.
- **Secure Code Generation System:** Developed SGCode, an LLM system for secure code generation with static analysis. Conducted analysis showing up to 90% functionality deviation on vulnerability patches by LLMs.

FPT Software | *Data Engineer*

November 2023 – June 2024

- **Risk Assessment QA system:** Developed an LLM-based QA system that uses Vector DB and RAG. Improved the cost, accuracy, and time of risk assessment process of a US banking client by 30%.

Productminds | *AI Engineer*

September 2023 – April 2024

- **Danish News Trend Analysis:** Built an NLP pipeline to automatically crawl Danish news articles and extract named entities, serving as the core data source for trend analysis products.

PUBLICATIONS

Papers

- [1] **Ton, Khiem*** and N. Phan, “Vrpo: Policy optimization for functionality-preserving automated vulnerability repair,” 2026, Under review at EMNLP 2026.
- [2] K. Nguyen*, **Ton, Khiem***, N. Phan, et al., “Noir: Privacy-preserving generation of code with open-source llms,” in *Proceedings of the 35th USENIX Security Symposium*, ser. **USENIX Security '26**, 2026.
- [3] **Ton, Khiem***, N. Nguyen*, M. Nazzal, et al., “Sgcode: A flexible prompt-optimizing system for secure generation of code,” ser. **CCS '24**, Association for Computing Machinery, 2024. DOI: 10.1145/3658644.3691367

Patents

- [1] N. Phan, K. Nguyen, **Ton, Khiem**, et al., *System and method for private generation of code*, International Patent Application, No. PCT/US25/042391, 2025.

TEACHING EXPERIENCE

- Teaching Assistant: Artificial Intelligence (FA25, SP26), Operating Systems Design (SU25), Big Data (SP25)
- Lecturer: Social Network Analysis (FA25)

SERVICE

- Volunteer: ICRA (2025)
- Reviewer: ICML (2026)